Math 1200 Lecture 3

Sep 10 2026

Topics: 13.1-13.2: Vector valued functions, derivatives and integrals of vector-valued fns

Name:_____

Collaborators:_____

Group work 1 Plot the space curves parametrized by the vector-valued functions
(a) $\vec{r}(t) = \langle \cos(t), \sin(t), e^t \rangle$ (b) $\vec{r}(t) = \langle \cos(4t), \sin(4t), e^{4t} \rangle$ (c) $\vec{r}(t) = \langle \cos(t), \sin(t), \log(t) \rangle$

Group work 2 Parametrize the intersection of the plane $z = 2y + 3$ with the surface $z = x^2 + y^2$.[1]

[1] Hint: At some point you may need to complete the square!

Group work 3 Find the tangent vector and unit tangent vector at the point $(3, 3, 0)$ to the curve C parametrized by $\vec{r}(t) = \langle t^2 + 2t, 3t, \ln(t^2) \rangle$ for $t > 0$.

Group work 4 Suppose a space curve C passes through the point $(1, 1, 1)$ and is parametrized by a function $\vec{r}(t)$ such that the tangent vector to C at time t is $\langle 1, t, t^2 \rangle$. Parametrize C.